

C1 final practice

1. (Spring 2022, #8) A car is driving on the road and suddenly notices an obstacle 500 feet away and immediately starts braking. The car's velocity in ft/s is measured every four seconds and is given in the table below

t	0	4	8	12	16	20
v	60	50	20	12	4	0

- (a) Roughly at what time does the car come to a stop? Justify your answer.

At $t = 20$ because $v(20) = 0$.

- (b) Give an upper and a lower estimate for the distance traveled between $t = 0$ and $t = 20$ seconds. Justify your answer.

To get an upper estimate, we should use the largest value in each sub interval.

Upper estimate: $4 \times (60 + 50 + 20 + 12 + 4) = 4 \times 146 = 584$

To get a lower estimate, we should use the smallest value in each sub interval.

Upper estimate: $4 \times (50 + 20 + 12 + 4 + 0) = 4 \times 86 = 344$

- (c) Based on your results, which of the following statements can you make? Please explain your answer briefly below.

- The car hits the obstacle.
- It does not hit the obstacle.
- We can not tell from just the given data.

The actual distance lies on the interval $[344, 584]$

But we can not say if it is greater or less than 500. So, we can not tell from just the given data.

2. (Fall 2020, #1) Write the following limit as a definite integral. You may use the properties of logarithms to help with your algebra. Do not evaluate the integral.

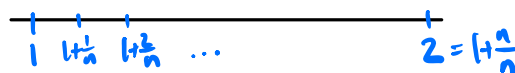
$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right)^3 \left(1 + \frac{2}{n}\right)^3 \cdots \left(1 + \frac{n}{n}\right)^3 \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \left[\ln \left(\left(1 + \frac{1}{n}\right)^3 \right) + \ln \left(\left(1 + \frac{2}{n}\right)^3 \right) + \dots + \ln \left(\left(1 + \frac{n}{n}\right)^3 \right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{i=1}^n \ln \left(\left(1 + \frac{i}{n}\right)^3 \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n 3 \ln \left(1 + \frac{i}{n} \right) \cdot \frac{1}{n}$$

$$= \int_1^2 3 \cdot \ln(x) dx$$



3. (Spring 2020, #6) Consider the integral $\int_0^1 \sqrt{1+x^2} dx$.

$$f(x) = \sqrt{1+x^2}$$

- (a) Express the integral as a left Riemann sum with 4 sub-intervals of equal width. You may leave your answer as an unsimplified sum.

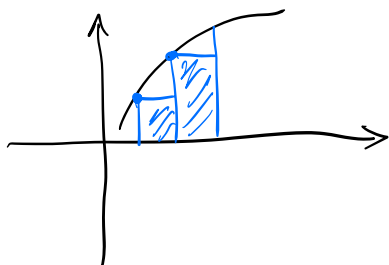
Define $f(x) = \sqrt{1+x^2}$.

$$\Delta x = \frac{1-0}{4} = \frac{1}{4} \Rightarrow \text{sub intervals: } [0, \frac{1}{4}], [\frac{1}{4}, \frac{2}{4}], [\frac{2}{4}, \frac{3}{4}], [\frac{3}{4}, \frac{4}{4}]$$

$$\begin{aligned} \text{Left Riemann sum: } & \frac{1}{4} \times f(0) + \frac{1}{4} f\left(\frac{1}{4}\right) + \frac{1}{4} f\left(\frac{2}{4}\right) + \frac{1}{4} f\left(\frac{3}{4}\right) \\ &= \frac{1}{4} \cdot \sqrt{1+0^2} + \frac{1}{4} \sqrt{1+\left(\frac{1}{4}\right)^2} + \frac{1}{4} \sqrt{1+\left(\frac{2}{4}\right)^2} + \frac{1}{4} \sqrt{1+\left(\frac{3}{4}\right)^2} \\ &= \frac{1}{4} \left(1 + \sqrt{\frac{17}{16}} + \sqrt{\frac{20}{16}} + \sqrt{\frac{25}{16}} \right) \end{aligned}$$

- (b) Is the above Riemann sum in (a) greater than or less than the value of the integral? Briefly explain.

Since $f(x)$ is increasing $f(\text{"left endpoint"})$ or $f(x_i)$ is the lowest value on its corresponding interval $[f(x_i), f(x_{i+1})]$. So we are underestimating the area under the function on each interval. So the Riemann sum in (a) is less than the value of the integral.



Briefly:

Since the function is an increasing function, our estimation is less than the actual value.

4. (Fall 2019, #3) Consider the integrals $I = \int_0^2 (1+x^2) dx$ and $J = \int_0^2 e^{x^2} dx$.

Note: the value of the integral J cannot be evaluated explicitly.

- (a) Write I as the limit of a right Riemann sum.

I want to consider n subintervals on the interval $[0, 2]$. So, $\Delta x = \frac{2-0}{n} = \frac{2}{n}$.

Thus: $x_0 = 0$, $x_1 = \frac{2}{n}$, $x_2 = \frac{4}{n}$, $\dots$, $x_n = \frac{2n}{n}$. Or generally: $x_i = \frac{2i}{n}$ for $i = 0, \dots, n$.

Define $f(x) = (1+x^2)$. The right Riemann sum is:

$$\sum_{i=1}^n \Delta x f(x_i) = \sum_{i=1}^n \frac{2}{n} \left(1 + \left(\frac{2i}{n}\right)^2\right) = \frac{2}{n} \sum_{i=1}^n \left(1 + \left(\frac{2i}{n}\right)^2\right)$$

- (b) Compute the limit of the Riemann sum. **Hint:** $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

We need to simplify the Riemann sum in part (a).

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \left(1 + \left(\frac{2i}{n}\right)^2\right) &= \lim_{n \rightarrow \infty} \frac{2}{n} \left(\underbrace{\sum_{i=1}^n 1}_n + \sum_{i=1}^n \left(\frac{2i}{n}\right)^2 \right) = \lim_{n \rightarrow \infty} \frac{2}{n} \times n + \frac{2}{n} \sum_{i=1}^n \left(\frac{2i}{n}\right)^2 \\ &= \lim_{n \rightarrow \infty} 2 + \frac{2}{n} \sum_{i=1}^n \frac{4i^2}{n^2} = \lim_{n \rightarrow \infty} 2 + \frac{2}{n} \times \frac{4}{n^2} \sum_{i=1}^n i^2 = \lim_{n \rightarrow \infty} 2 + \frac{8}{n^2} \sum_{i=1}^n i^2 = \lim_{n \rightarrow \infty} 2 + \frac{8}{n^2} \left(\frac{n(n+1)(2n+1)}{6} \right) \\ &= \lim_{n \rightarrow \infty} 2 + \frac{8}{n^2} \times \frac{2n^3 + 3n^2 + n}{6} = \lim_{n \rightarrow \infty} 2 + \frac{16n^3 + 24n^2 + 8n}{6n^2} = \lim_{n \rightarrow \infty} 2 + \frac{16n}{6} = 2 + \frac{16}{6} \approx 4.66 \end{aligned}$$

- (c) Given that $e^{x^2} \geq 1+x^2$ for all values of x , determine which value is larger: J or 4.59?

$$\text{Since } e^{x^2} \geq 1+x^2 \text{ for all values of } x \Rightarrow \int_0^2 e^{x^2} dx \geq \int_0^2 (1+x^2) dx$$

$$\Rightarrow J \geq I \approx 4.66$$

So J is larger than 4.59.

5. (Fall 2006, #8)

(a) Write the following limit as an integral.

$$\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \sqrt{4 - \left(\frac{2i}{n}\right)^2}$$

$\uparrow$ Δx $\uparrow$ x_i

By looking at the (right) Riemann sum, we can find Δx and x_i 's and $f(x)$.

$$\Delta x = \frac{2}{n}, \quad x_i = \frac{2i}{n}, \quad f(x) = \sqrt{4 - x^2}$$

$$x_0 = \frac{2 \cdot 0}{n} = 0 \quad \text{or we can find } x_0 \text{ using } x_1 \text{ and } \Delta x : x_0 = x_1 - \Delta x = \frac{2 \cdot 1}{n} - \frac{2}{n} = 0$$

$$x_n = \frac{2n}{n} = 2$$

Thus the limits of the corresponding definite integral are 0 and 2.

$$\int_0^2 \sqrt{4 - x^2} dx$$

(b) Compute the limit by interpreting the integral as an area.

We have $y = \sqrt{4 - x^2}$ for $0 \leq x \leq 2$. Since $y = \sqrt{4 - x^2}$, y is also positive.

Thus x and y are positive.

$y = \sqrt{4 - x^2} \Rightarrow y^2 = 4 - x^2 \Rightarrow x^2 + y^2 = 2^2$: a circle with the center at $(0,0)$ and radius 2.

But we should only consider $x \geq 0$ and $y \geq 0$.

